

Solution Code - Module 3-Core Java - StringReversal

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q15_StringReversal.java

```
1 | import java.util.Scanner;
2 |
3 | public class Q15_StringReversal {
4 |     public static void main(String[] args) {
5 |         Scanner scanner = new Scanner(System.in);
6 |         System.out.print("Enter a string to reverse: ");
7 |         if (scanner.hasNextLine()) {
8 |             String input = scanner.nextLine();
9 |
10 |             // Method 1: Using StringBuilder
11 |             String reversedBuilder = new StringBuilder(input).reverse().toString();
12 |
13 |             // Method 2: Using a loop (for verification of manual loop logic)
14 |             char[] charArray = input.toCharArray();
15 |             StringBuilder reversedLoop = new StringBuilder();
16 |             for (int i = charArray.length - 1; i >= 0; i--) {
17 |                 reversedLoop.append(charArray[i]);
18 |             }
19 |
20 |             System.out.println("Reversed string (using StringBuilder): " + reversedBuilder);
21 |             System.out.println("Reversed string (using manual loop): " + reversedLoop.toString());
22 |         } else {
23 |             System.out.println("Error: No input found.");
24 |         }
25 |         scanner.close();
26 |     }
27 | }
```